

Stormwater Regulation Summary: Columbus, OH

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary drivers: MS4 permit compliance, Scioto River and Big Walnut Creek water quality.
- Columbus Stormwater Drainage Manual + Ohio EPA CGP govern stormwater management.
- City of Columbus Department of Public Utilities reviews all stormwater management plans.
- Franklin County also has jurisdiction over unincorporated areas with separate requirements.
- Columbus Climate Action Plan (2021) encourages green infrastructure and renewable energy adoption.

2. Typical Soil Conditions

- Columbus sits on glacial till plains — predominantly heavy clay soils (HSG C-D).
- Typical soils: Crosby silt loam, Kokomo silty clay loam — poor natural drainage.
- Hydrologic Soil Group: Predominantly C-D across the metro area.
- Scioto River floodplain has alluvial soils with variable drainage — some areas better for infiltration.
- Implication: Clay soils limit infiltration in most of Columbus — retention and controlled release are the practical approach.
- *Eastern suburbs (Gahanna, Reynoldsburg) may have slightly better drainage — site-specific testing required.*

3. Green Roof Requirements

- No green roof mandate in Columbus — adoption is voluntary and incentive-based.
- Columbus accepts green roofs as an approved post-construction BMP in their Stormwater Drainage Manual.
- Columbus Green Business Program provides recognition and some incentives for sustainable building practices.
- Ohio PACE financing available for green roof and solar installations.
- OSU campus and Nationwide Children's Hospital have institutional sustainability mandates that drive adoption.
- Short North and Franklinton redevelopment projects on constrained lots benefit from rooftop BMPs.

4. TSS Requirements

- Columbus follows Ohio EPA 80% TSS removal requirement for post-construction stormwater.
- Green roofs receive TSS credit for captured volume — consistent with Ohio EPA BMP manual.
- The 0.90" WQv capture addresses the first-flush pollutant load where TSS concentrations are highest.
- Ground-level runoff from parking and roadways may require additional treatment beyond green roof.
- *Confirm current TSS credit methodology in Columbus Stormwater Drainage Manual — may have been updated.*

5. Nature-Based Retention Requirements

- WQv: 0.90" from all developed impervious area (per Ohio EPA CGP standard).
- Critical Storm Method: Size detention to control the storm event whose post-development peak most exceeds pre-development.
- CN (Curve Number) method is the standard design approach, per NRCS TR-55.
- Typical Columbus urban CN: 85-98 (developed); with green roof: 65-75.
- Columbus may accept WQv management through retention (infiltration, evapotranspiration) or treatment (filtration).

6. Allowable Outflow Rates

- Critical Storm Method: Post-development peak $\leq$ pre-development for the critical storm (typically near 1-year event).
- 1-year through 100-year storms: peak flow control to pre-development rates.
- Columbus does not specify a flat l/s/ha rate — control is relative to pre-development.
- For reference, typical pre-development Columbus site (CN 70): approximately 18-28 l/s/ha for the 10-year storm.
- *Verify current standards in Columbus Stormwater Drainage Manual — requirements may differ by watershed.*

Items in *italic* should be verified against current municipal codes. | Generated by Sempergreen USA Stormwater BMP Comparison Tool